

Ex 5b

1. Aim

To perform exploratory data analysis, data pre-processing, feature engineering, and graphical visualization on the Online Retail transaction dataset using Python libraries (pandas, matplotlib, and seaborn).

2. Software / Tools Required

- **Language:** Python 3.x
- **Libraries:** pandas, numpy, matplotlib.pyplot, seaborn
- **Environment:** Jupyter Notebook / Google Colab / VS Code

3. Algorithm / Steps

1. **Load Dataset:** Import required modules and load OnlineRetail.csv using appropriate encoding.
2. **Data Cleaning:**
 - Check structural metadata (info(), describe()).
 - Handle missing values in CustomerID and Description.
 - Filter out cancelled transactions (invoices starting with 'C') and non-positive quantity/price entries.
3. **Feature Engineering:** Compute total sales ($\text{TotalSales} = \text{Quantity} * \text{UnitPrice}$) and extract date components (Month, DayOfWeek).
4. **Data Visualization:**
 - Plot distribution of product prices and order quantities.
 - Visualize top-selling products and top countries by total revenue.
 - Generate monthly revenue trends and scatter plots showing price vs. quantity relationships.
5. **Output Generation:** Render plots with appropriate axis titles, legends, and grid lines.

Program:

```

import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# 1. Load Dataset & Feature Engineering
df = pd.read_csv("OnlineRetail.csv", encoding="ISO-8859-1")
df["TotalSales"] = df["Quantity"] * df["UnitPrice"]
df["IsCancelled"] = df["InvoiceNo"].str.startswith("c", na=False).astype(int)

# Filtered/sample subset for clean visualization (removes extreme retail outlier)
sample_df = df[
    (df["Quantity"] > 0)
    & (df["UnitPrice"] > 0)
    & (df["Quantity"] <= 50)
    & (df["UnitPrice"] <= 20)
].sample(2000, random_state=42)

print(df.head())

```

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Python

```

...
InvoiceNo StockCode Description Quantity \
0 536365 85123A WHITE HANGING HEART T-LIGHT HOLDER 6
1 536365 71053 WHITE METAL LANTERN 6
2 536365 84406B CREAM CUPID HEARTS COAT HANGER 8
3 536365 84029G KNITTED UNION FLAG HOT WATER BOTTLE 6
4 536365 84029E RED WOOLLY HOTTIE WHITE HEART. 6

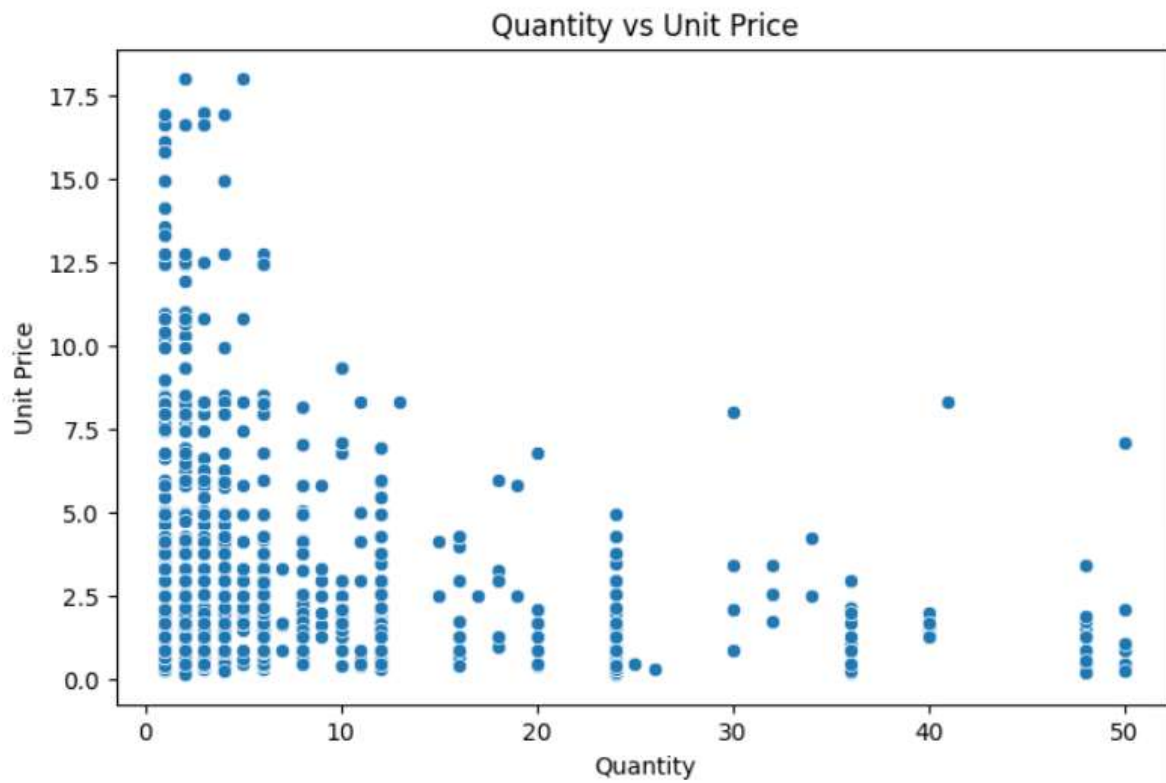
InvoiceDate UnitPrice CustomerID Country TotalSales \
0 12/1/2010 8:26 2.55 17850.0 United Kingdom 15.30
1 12/1/2010 8:26 3.39 17850.0 United Kingdom 20.34
2 12/1/2010 8:26 2.75 17850.0 United Kingdom 22.00
3 12/1/2010 8:26 3.39 17850.0 United Kingdom 20.34
4 12/1/2010 8:26 3.39 17850.0 United Kingdom 20.34

IsCancelled
0 0
1 0
2 0

```

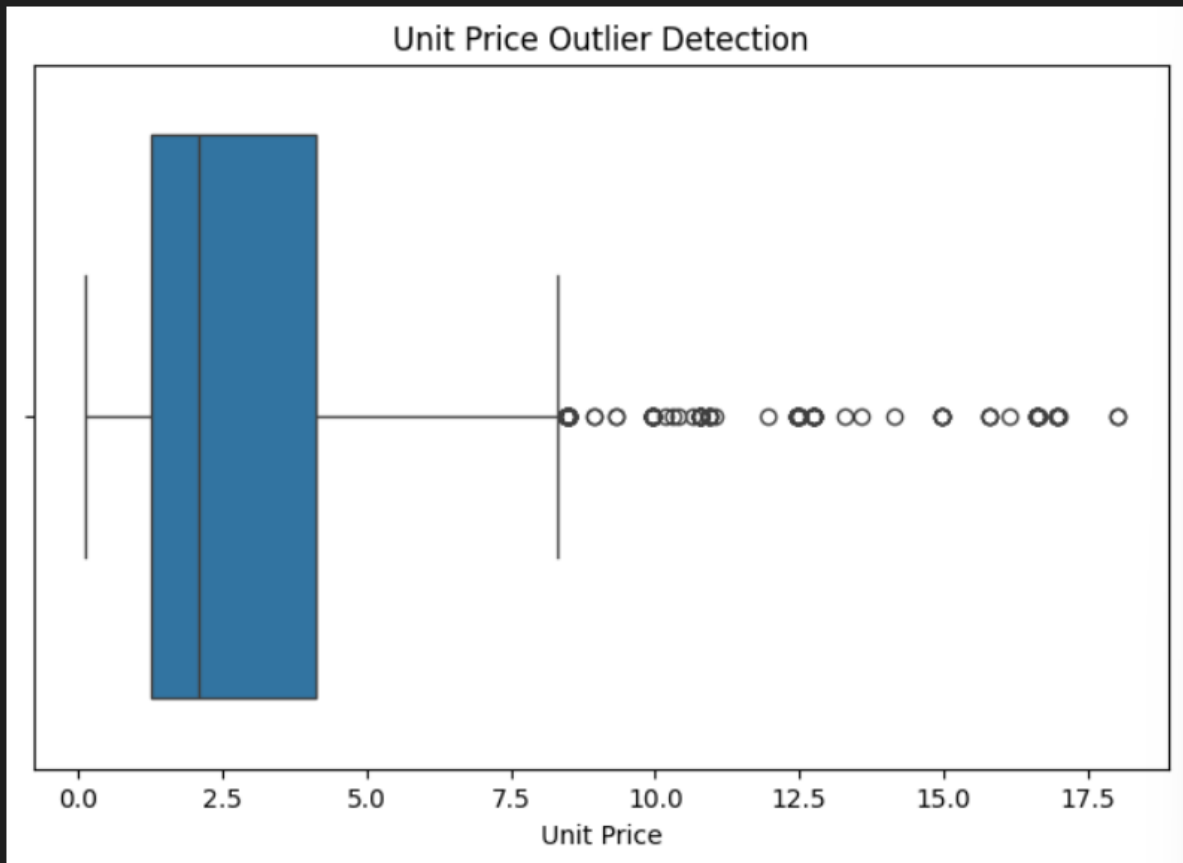
```
# 2. Scatter Plot: Quantity vs Unit Price
plt.figure(figsize=(8, 5))
sns.scatterplot(data=sample_df, x="Quantity", y="UnitPrice")
plt.title("Quantity vs Unit Price")
plt.xlabel("Quantity")
plt.ylabel("Unit Price")
plt.show()
```

Python



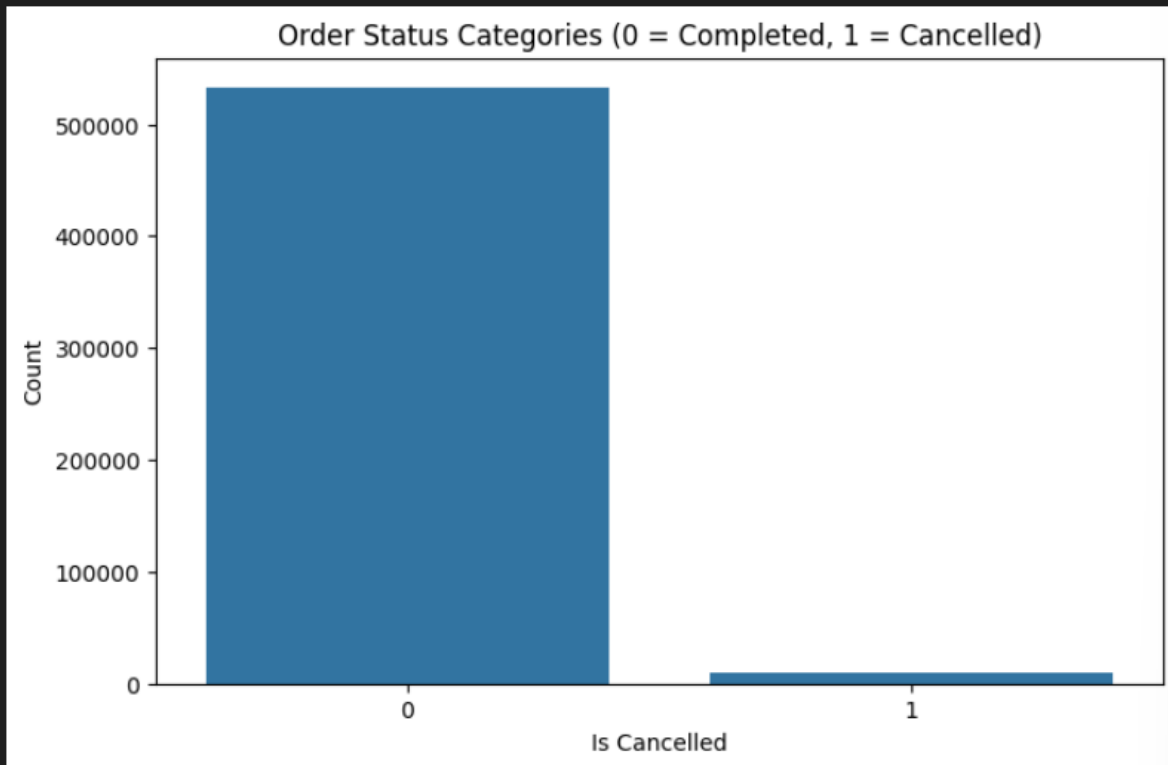
```
plt.figure(figsize=(8, 5))
sns.boxplot(x=sample_df["UnitPrice"])
plt.title("Unit Price Outlier Detection")
plt.xlabel("Unit Price")
plt.show()
```

Python



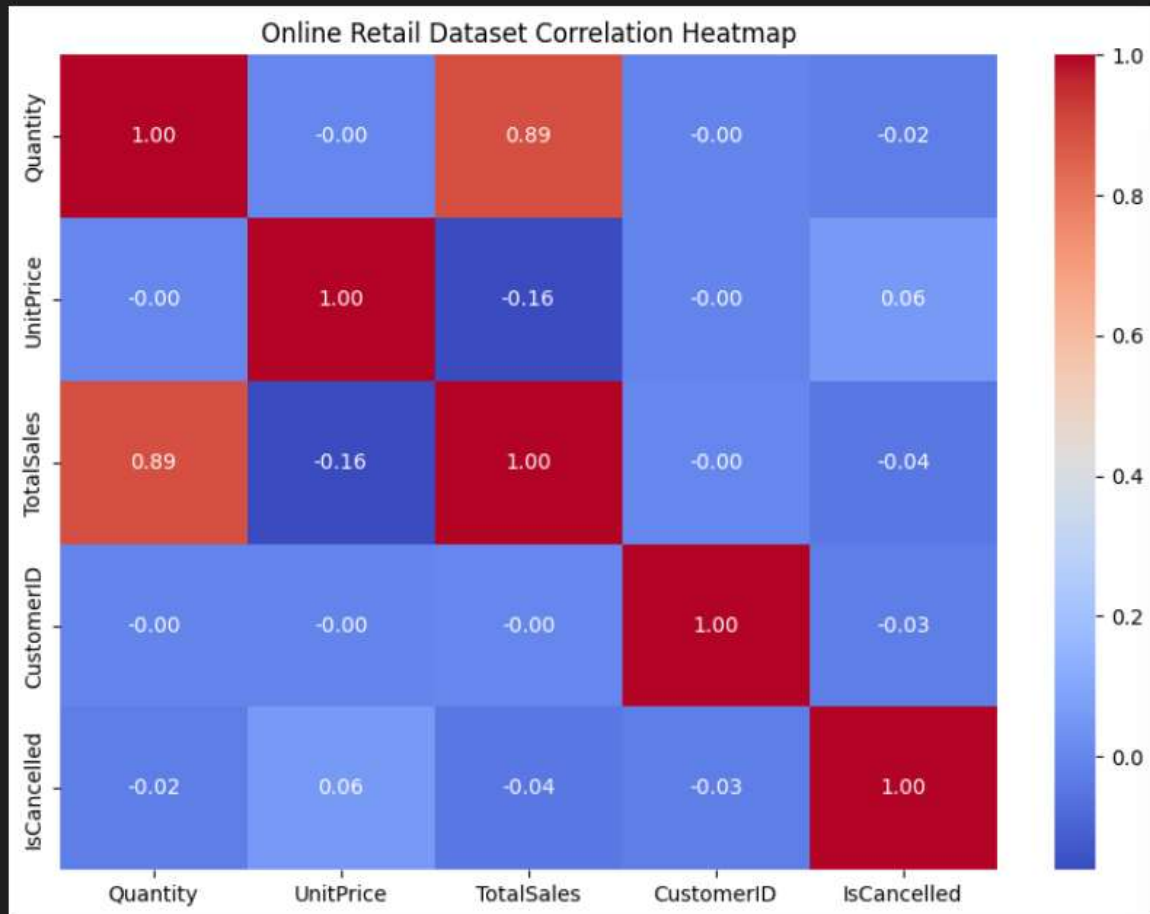
```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="IsCancelled")
plt.title("Order Status Categories (0 = Completed, 1 = Cancelled)")
plt.xlabel("Is Cancelled")
plt.ylabel("Count")
plt.show()
```

Python



```
plt.figure(figsize=(10, 7))
corr = df[["Quantity", "UnitPrice", "TotalSales", "CustomerID", "IsCancelled"]].
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Online Retail Dataset Correlation Heatmap")
plt.show()
```

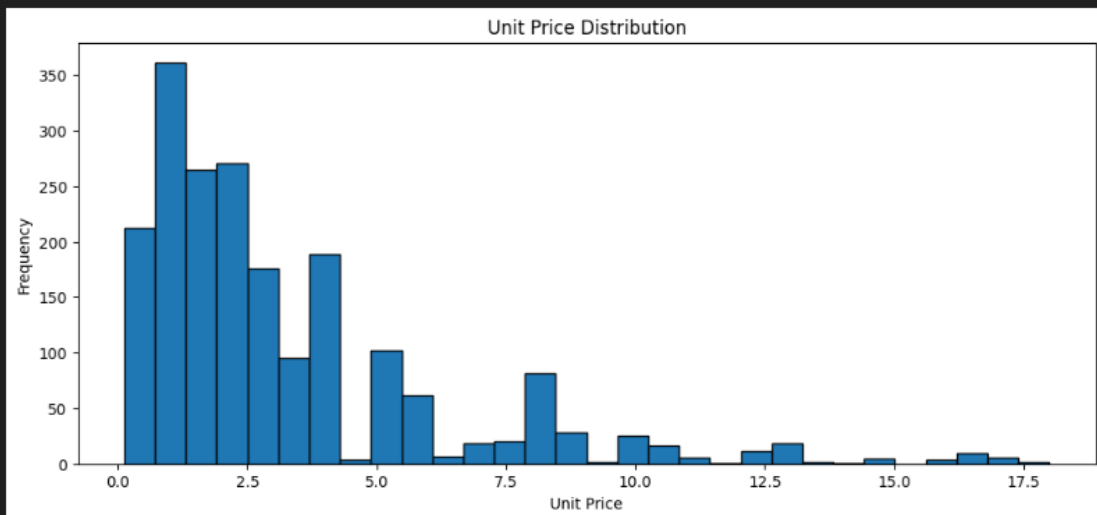
Python



```
plt.figure(figsize=(12, 5))
plt.hist(sample_df["UnitPrice"], bins=30, edgecolor="black")
plt.title("Unit Price Distribution")
plt.xlabel("Unit Price")
plt.ylabel("Frequency")
plt.show()
```

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Python

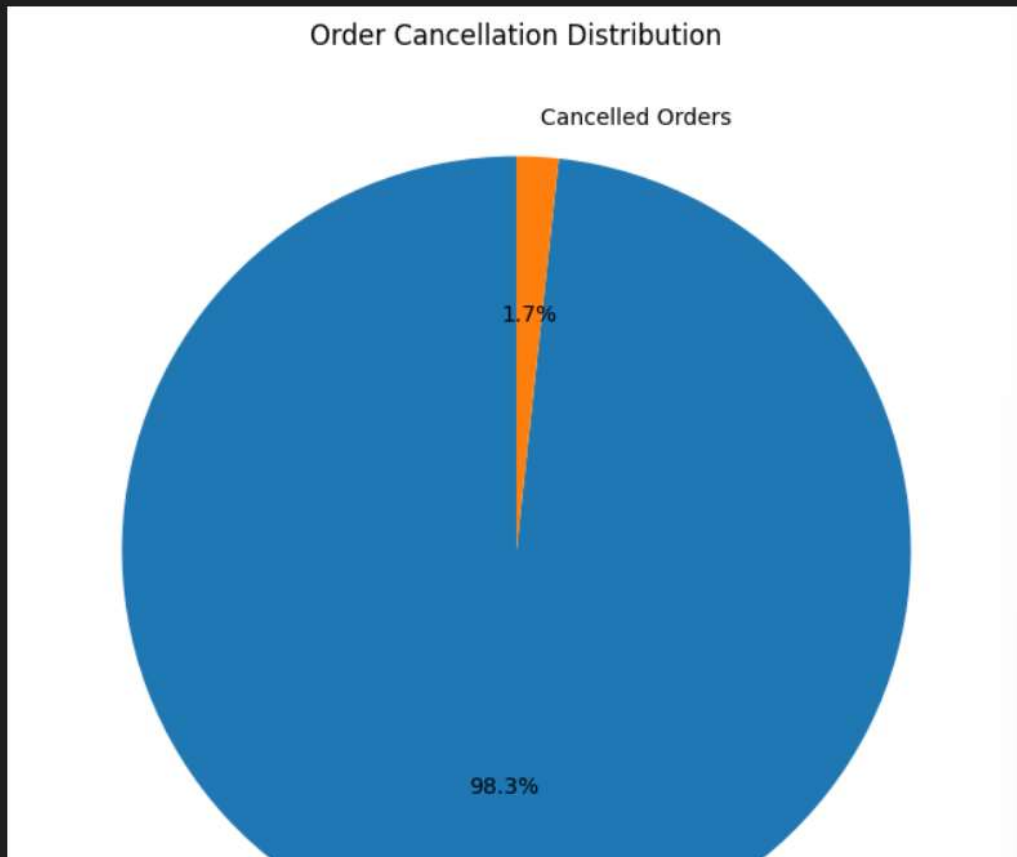


```
cancellation_counts = df["IsCancelled"].value_counts()
plt.figure(figsize=(8, 8))
plt.pie(
    cancellation_counts,
    labels=["Completed Orders", "Cancelled Orders"],
    autopct="%1.1f%%",
    startangle=90,
)
plt.title("Order Cancellation Distribution")
plt.show()
```

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Python

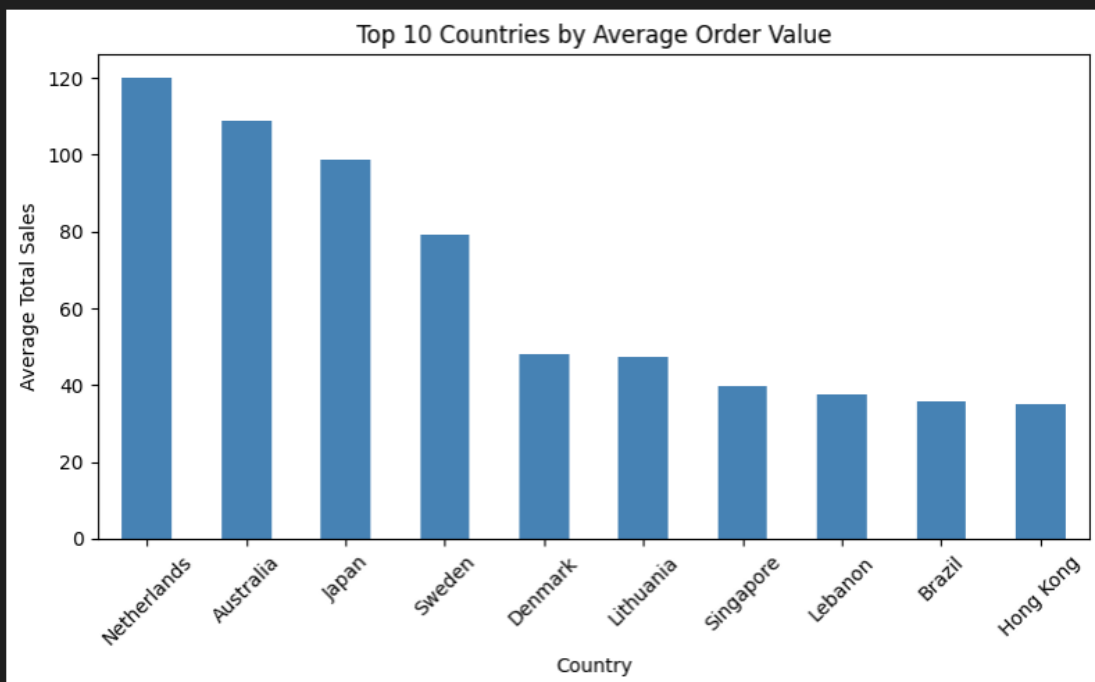
...



```
top_countries = (
    df.groupby("Country")["TotalSales"]
      .mean()
      .sort_values(ascending=False)
      .head(10)
)
plt.figure(figsize=(8, 5))
top_countries.plot(kind="bar", color="steelblue")
plt.title("Top 10 Countries by Average Order Value")
plt.ylabel("Average Total Sales")
plt.xlabel("Country")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

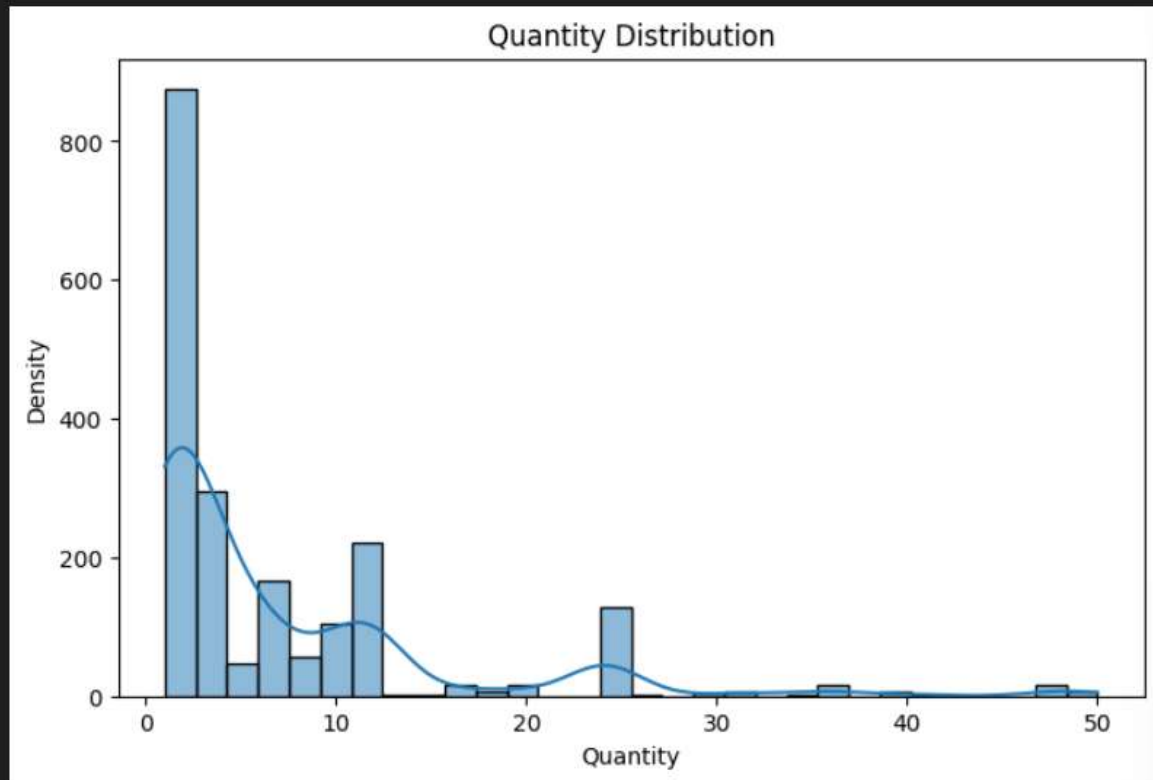
[12]

Python



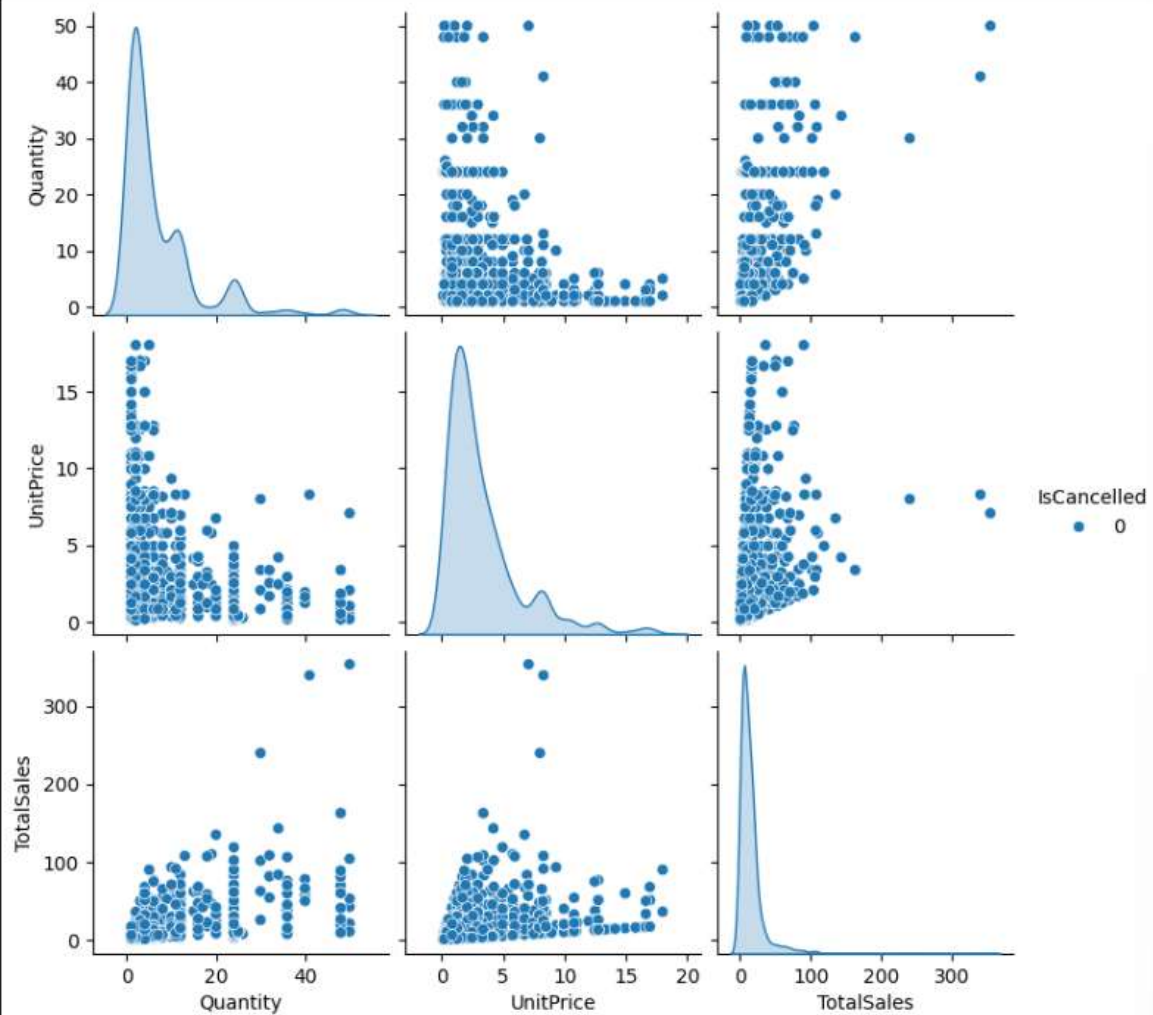
```
plt.figure(figsize=(8, 5))
sns.histplot(sample_df["Quantity"], bins=30, kde=True)
plt.title("Quantity Distribution")
plt.xlabel("Quantity")
plt.ylabel("Density")
plt.show()
```

Python



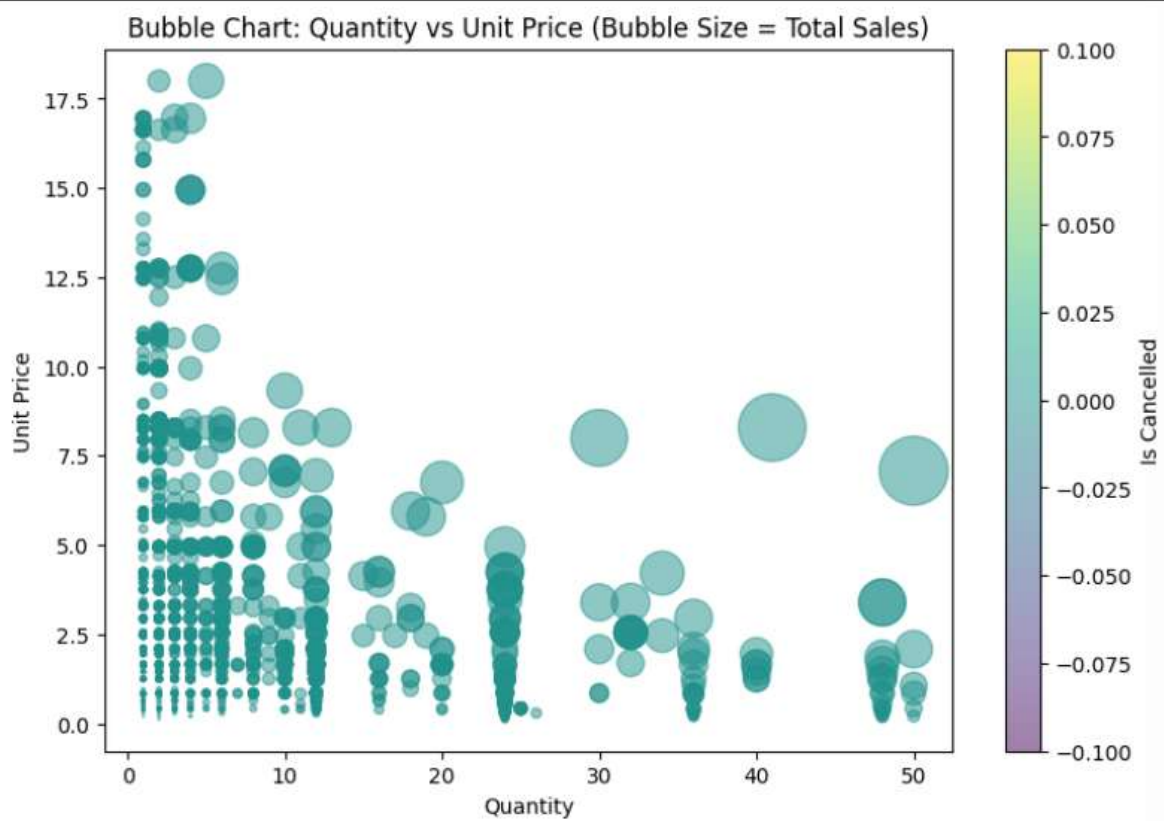
```
sns.pairplot(
    sample_df[["Quantity", "UnitPrice", "TotalSales", "IsCancelled"]],
    hue="IsCancelled",
)
plt.show()
```

Python



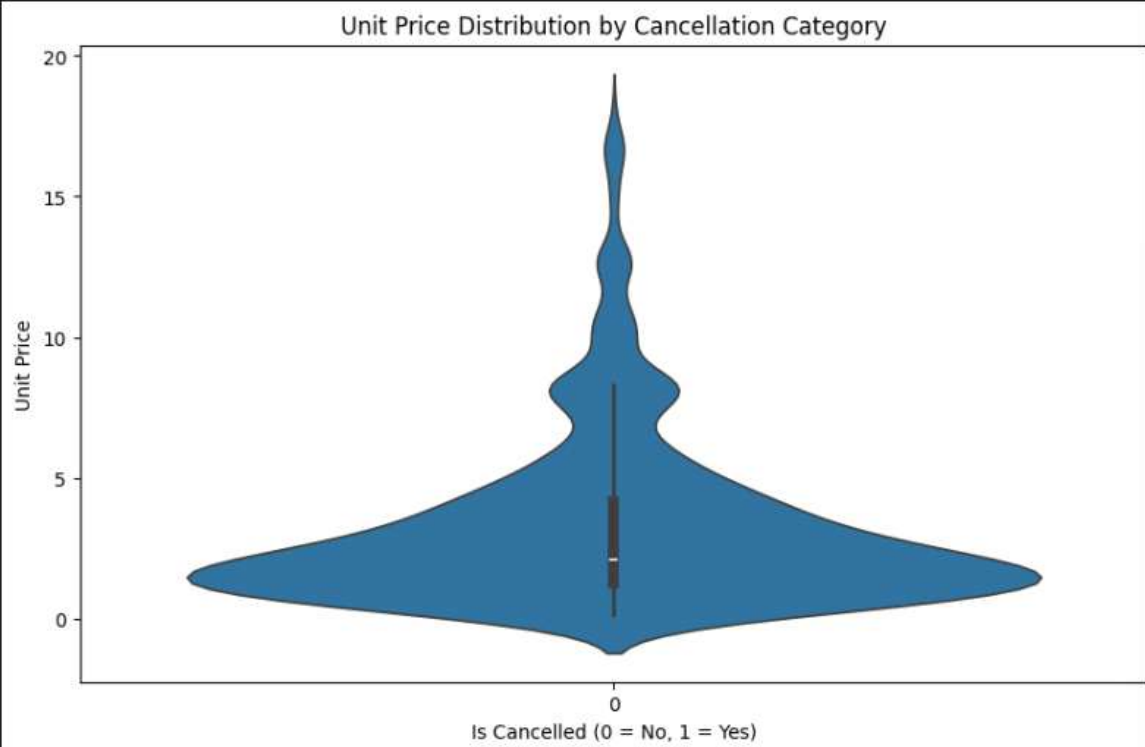
```
plt.figure(figsize=(9, 6))
plt.scatter(
    sample_df["Quantity"],
    sample_df["UnitPrice"],
    s=sample_df["TotalSales"] * 3,
    alpha=0.5,
    c=sample_df["IsCancelled"],
    cmap="viridis",
)
plt.colorbar(label="Is Cancelled")
plt.xlabel("Quantity")
plt.ylabel("Unit Price")
plt.title("Bubble Chart: Quantity vs Unit Price (Bubble Size = Total Sales)")
plt.show()
```

Python



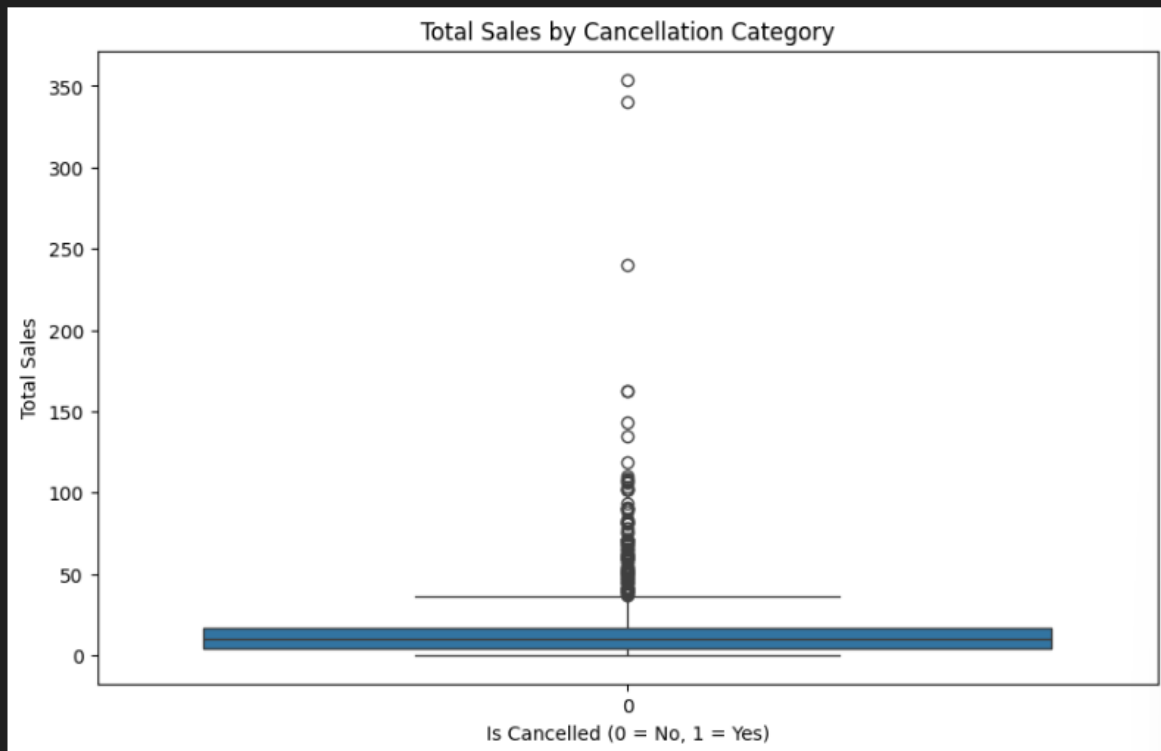
```
plt.figure(figsize=(10, 6))
sns.violinplot(data=sample_df, x="IsCancelled", y="UnitPrice")
plt.title("Unit Price Distribution by Cancellation Category")
plt.xlabel("Is Cancelled (0 = No, 1 = Yes)")
plt.ylabel("Unit Price")
plt.show()
```

Python



```
plt.figure(figsize=(10, 6))
sns.boxplot(data=sample_df, x="IsCancelled", y="TotalSales")
plt.title("Total Sales by Cancellation Category")
plt.xlabel("Is Cancelled (0 = No, 1 = Yes)")
plt.ylabel("Total Sales")
plt.show()
```

Python



Result

The exploratory data analysis and visualization of the online retail dataset were successfully completed. Key trends in sales, product demand, and international revenue were identified through clear graphical plots.